

ShadowEx Parameter Usage Guide (English)

ShadowEx is a custom shader extension for **lilToon** that enhances rendering capabilities by adding screen-space shadows (SSAO / Contact Shadows), multi-layer MatCaps, extra normal maps, stylized specular, and more.

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1. Shared FX Mask

A shared mask texture system using **2 RGBA textures (8 channels in total)** across multiple FX features (Rim Shade, MatCap Layers, Contact Shadow, Additional Specular, Extra Normals).

- **Properties:**
 - `_CustomFXMask` : **FX Mask 1 (RGBA)**
 - `_CustomFXMask2` : **FX Mask 2 (RGBA)**
 - **Features:**
 - Select the mask channel to use per feature (**Mask1 R/G/B/A** to **Mask2 R/G/B/A**) using the **Mask Channel** popup.
 - Only masks referenced by active features are sampled, keeping texture sampling overhead minimal (maximum 2 texture samples per pixel regardless of active FX count).
 - **Recommended Import Setting:** Set **sRGB (Color Texture)** to **OFF (Linear)** in texture import settings.
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2. SSAO (Screen Space Ambient Occlusion)

Calculates real-time ambient occlusion shadows in crevices and contact areas using the camera depth buffer (`_CameraDepthTexture`).

⚠ Requirement: Functions in environments like VRChat where `_CameraDepthTexture` is enabled (e.g., worlds with shadowed Directional Light).

- **Key Parameters:**
 - **Enable SSAO** (`_CustomSSAOEnabled`): Toggles SSAO ON/OFF.
 - **Occlusion Color** (`_CustomSSAOCOLOR`): Color of the ambient occlusion shadow (default: Black).

- **Strength** (`_CustomSSAStrength`) [Range: 0 - 8]: Intensity of occlusion.
- **Power** (`_CustomSSAPower`) [Range: 0.1 - 4]: Controls contrast and softness of gradation. Values below 1 soften the shadow gradient.
- **Sample Length (m)** (`_CustomSSASampleLength`) [Range: 0.001 - 1]: Search radius for occlusion in meters.
- **Min Distance (m) / Max Distance (m)** (`_CustomSSAMinDistance`, `_CustomSSAMaxDistance`): Distance range for occlusion evaluation.
- **Depth Bias (m)** (`_CustomSSABias`): Depth offset to prevent self-occlusion artifacts/flickering.
- **Fade Distance (m)** (`_CustomSSAFadeDistance`): Distance at which AO gradually fades out.
- **Blur** (`_CustomSSABlur`): Lightweight blur to smooth out noise and banding.
- **Quality** (`_CustomSSAQuality`) [1 - 4]: Sampling quality multiplier for 12-sample kernel.
- **Dither (IGN)** (`_CustomSSADither`): Toggles Interleaved Gradient Noise dithering jitter.

3. Contact Shadow

Raymarches the camera depth buffer towards the main light direction to cast crisp, short-range contact shadows near feet, clothing folds, etc.

⚠ Requirement: Requires a world with a shadowed Directional Light to function.

- **Key Parameters:**
 - **Enable Contact Shadow** (`_CustomContactShadowEnabled`): Toggles Contact Shadows ON/OFF.
 - **Shadow Color (Multiply)** (`_CustomContactShadowColor`): Multiplicative shadow color.
 - **Ray Length (m)** (`_CustomContactShadowLength`) [Range: 0.005 - 0.5]: Maximum raymarch length (intended for short distances of a few cm).
 - **Thickness (m)** (`_CustomContactShadowThickness`): Ray thickness considered as an occluder.
 - **Depth Bias (m)** (`_CustomContactShadowBias`): Offset to prevent self-shadowing artifacts.
 - **Blur** (`_CustomContactShadowBlur`): Softness of shadow edges.
 - **Blur Strength** (`_CustomContactShadowBlurStrength`): Scaling multiplier for distance/depth-based soft shadow softening.
 - **Quality** (`_CustomContactShadowQuality`) [1 - 4]: Step count quality multiplier (8 steps).
 - **Mask Channel** (`_CustomContactShadowMaskChannel`): Selects the channel from Shared FX Mask to limit shadow placement.

4. Extra Normals

Blends **2 additional independent tangent-space normal maps (1st / 2nd)** on top of lilToon's main normal map.

- **Key Parameters:**
 - **Enable Extra Normals** (`_CustomExtraNormalEnabled`): Toggles Extra Normals ON/OFF.
 - **Normal Map 1st / 2nd** (`_CustomExtraNormal1stTex`, `_CustomExtraNormal2ndTex`): Normal map textures (set Texture Type to `Normal map`).
 - **Strength** (`_CustomExtraNormalStrengthA`, `_CustomExtraNormalStrengthB`) [Range: 0 - 10]: Normal bump intensity scale.
 - **Tiling** (`_CustomExtraNormal1stScale`, `_CustomExtraNormal2ndScale`): Independent UV tiling scales for 1st and 2nd layers.

- **Mask Channel** (`_CustomExtraNormal1stMaskChannel`, `_CustomExtraNormal2ndMaskChannel`): Mask channel from Shared FX Mask to scale bump strength.
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5. Rim Light 2nd

An additional rim light layer applied during the base pass emission stage. Features both traditional Fresnel mode and silhouette-detecting **Depth Contour mode**.

- **Mode Selection** (`_CustomRim2ndMode`):
 - **0: Fresnel**: Standard view-angle rim highlight (zero extra samples).
 - **1: Depth Contour**: Detects silhouette edges from the depth buffer for uniform outline rim lighting.
 - **Common Parameters**:
 - **Rim Color (HDR)** (`_CustomRim2ndColor`): Emission color for the rim light.
 - **Main Color Strength** (`_CustomRim2ndMainStrength`): Blend ratio of main texture color into rim light color.
 - **Enable Lighting** (`_CustomRim2ndEnableLighting`): 0 = Constant color, 1 = Follows world light color & brightness.
 - **Shadow Mask** (`_CustomRim2ndShadowMask`): Attenuates rim light intensity in shadowed regions.
 - **Fresnel Mode Parameters**:
 - **Power** (`_CustomRim2ndPower`) / **Border** (`_CustomRim2ndBorder`) / **Blur** (`_CustomRim2ndBlur`)
 - **Depth Contour Mode Parameters**:
 - **Width (px)** (`_CustomRim2ndDepthWidth`): Line width of contour rim in pixels.
 - **Threshold (m)** (`_CustomRim2ndDepthThreshold`): Depth difference threshold for silhouette detection.
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6. Rim Shade

A simple multiplicative rim shade that darkens silhouette borders based on view and normal angles.

- **Features**:
 - Head-vector based calculation reduces VR eye parity differences.
 - Supported in Lite shaders. Applied prior to emission in the base pass so emission brightness is preserved.
 - **Key Parameters**:
 - **Shade Color (Multiply)** (`_CustomRimShadeColor`): Multiplicative darkening color.
 - **Border** (`_CustomRimShadeBorder`): Start position of border shading.
 - **Blur** (`_CustomRimShadeBlur`): Softness of the shade edge.
 - **Fresnel Power** (`_CustomRimShadeFresnelPower`): Falloff power exponent.
 - **Mask Channel** (`_CustomRimShadeMaskChannel`): Shared FX Mask channel for area control.
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7. Additional Specular

Adds stylized Blinn-Phong highlights relative to the main light direction for hair, skin, or fabric shine.

- **Key Parameters:**
 - **Specular Color (HDR) (`_CustomSpecColor`):** Highlight color.
 - **Smoothness (`_CustomSpecSmoothness`):** Controls highlight sharpness (0 = wide soft glow, 1 = sharp highlight).
 - **Strength (`_CustomSpecStrength`):** Specular intensity scale.
 - **Blend Mode (`_CustomSpecBlendMode`):**
 - `Normal` / `Add` / `Screen` / `Multiply`
 - **Enable Lighting (`_CustomSpecEnableLighting`):** Follows main light color when enabled.
 - **Shadow Mask (`_CustomSpecShadowMask`):** Suppresses specular highlights in shadow areas.
 - **Mask Channel (`_CustomSpecMaskChannel`):** Shared FX Mask channel for specular range.
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8. MatCap Layers (Layers 1-3)

Applies **up to 3 additional MatCap layers** on top of lilToon's primary MatCaps (1st/2nd).

- **Features:**
 - Reuses lilToon MatCap UVs (`fd.uvMat`) and texture samplers for low performance overhead.
 - Use black-background MatCaps for `Add` / `Screen`, and white-background MatCaps for `Multiply`.
 - **Per-Layer Parameters:**
 - **Enable / MatCap Texture / Color (HDR)**
 - **Blend Mode:** `Normal` / `Add` / `Screen` / `Multiply`
 - **Enable Lighting:** Toggles light color influence.
 - **Shadow Mask:** Attenuates MatCap brightness in shadow regions.
 - **Mask Channel:** Shared FX Mask channel selector.
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9. FX Mask Packer (Editor Tool)

An editor tool accessible via Unity menu: `Tools > dennokoworks > ShadowEx > FX Mask Packer`.

- **Features:**
 - Packs 1 to 4 grayscale/color mask images into a single RGBA PNG texture.
 - Automatically rescales input images to the maximum resolution using bilinear filtering if dimensions differ.
 - **Base Texture (Optional):** Specifies an existing packed texture to overwrite only selected channels while preserving unassigned channels.